SLIIT UNI
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ME 1060: DYNAMICS

Lecture 4: Kinetics of a Particle:

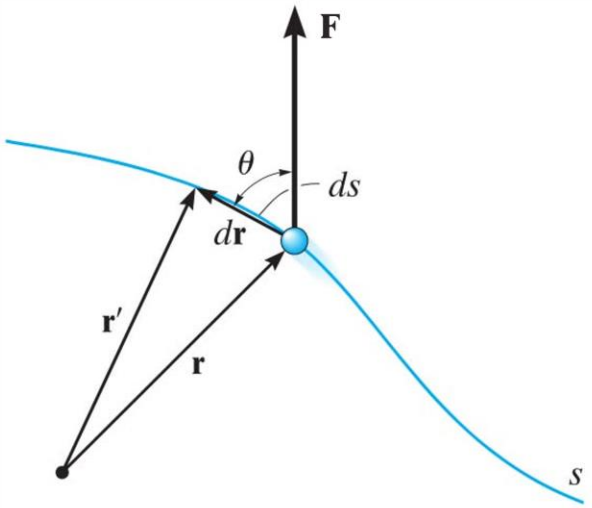
Work and Energy

Overview

- Work and Energy
- Principle of work and energy
- Power
- Efficiency.

Work and Energy

The Work of a Force



Consider a particle of mass m moving along the curved path.

F = resultant ΣF of all forces acting on the particle.

The position of m is established by the vector r, and the work dU done by F during the displacement dr

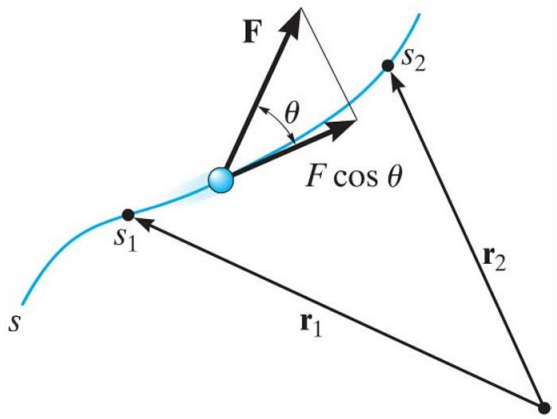
$$dU = \boldsymbol{F}.\boldsymbol{dr} = \mathrm{F}\ \mathrm{ds}\cos\theta$$

This expression represents the magnitude $(F\cos\theta).ds$, S is the scalar distance measured along the curve.

- F will do work on a particle only when the particle undergoes a displacement in the direction of the force.
- F is a scalar quantity

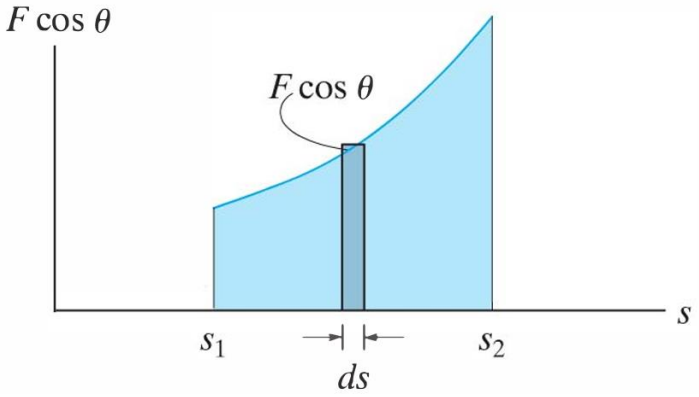
Work and Energy

The Work of a Variable Force



The work done by ***F*** during a finite movement of the particle from points **1** to **2** is

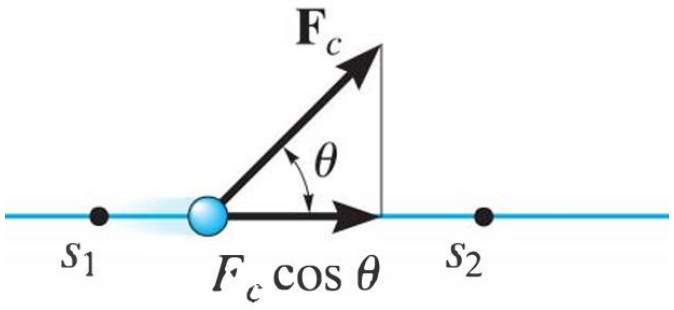
$$U_{1-2} = \int F.dr = \int_{S_1}^{S_2} (F\cos\theta)ds$$



The area under this graph bounded by S1 and S2 represents the total work

Work and Energy

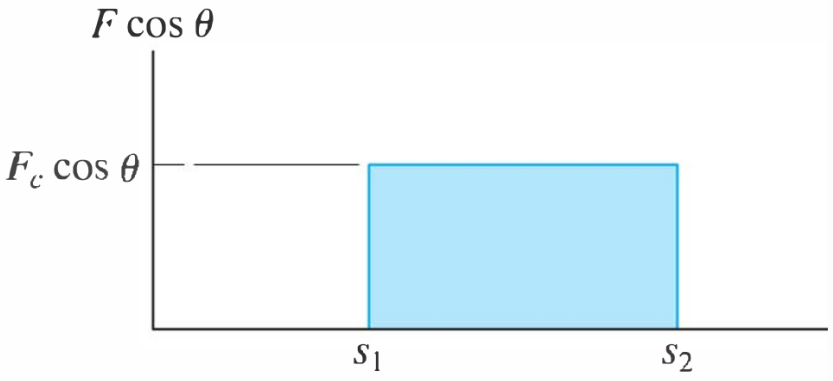
The Work of a Constant Force Moving Along a Straight Line



If the force F_c has a constant magnitude and acts at a constant angle θ from its straight-line path, the work done by F_c,

$$U_{1-2} = F\cos\theta \int_{S_1}^{S_2} ds$$

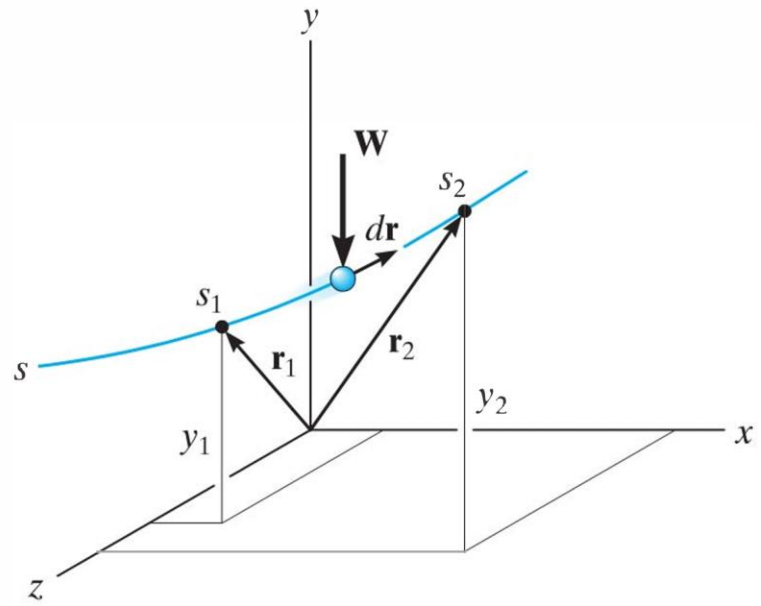
$$U_{1-2} = F\cos\theta\,(s_2 - s_1)$$



The work of F_c represent the area of the rectangle.

Work and Energy

The Work of a Weight



Consider a particle of weight W, which moves up along the path s from position s_1 to position s_2

$$U_{1-2} = \int F\, dr = \int_{r_1}^{r_2} (-W\boldsymbol{j})(dx\boldsymbol{i} + dy\boldsymbol{j} + dz\boldsymbol{k})$$

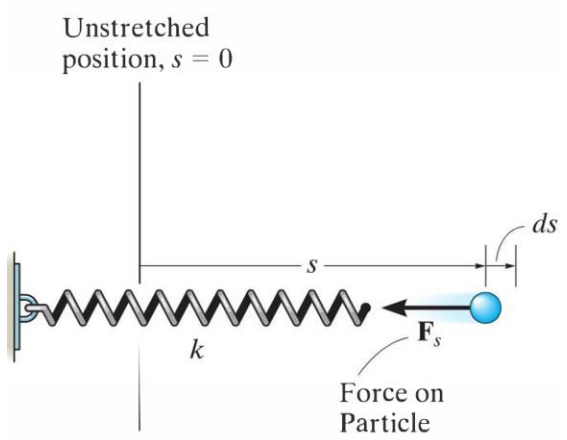
$$= \int_{y_1}^{y_2} -W\, dy = -W(y_2 - y_1)$$

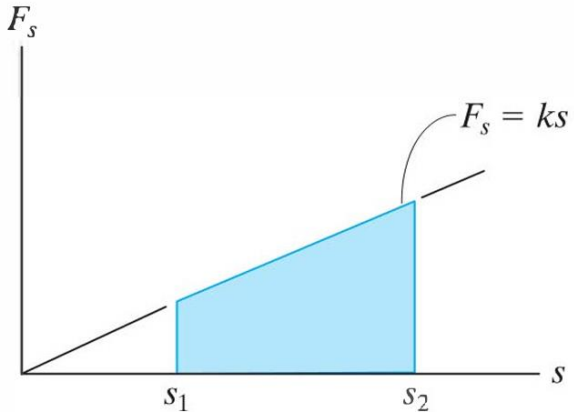
$$= -W\Delta y \qquad (\text{Where } W = mg)$$

Thus, the work is independent of the path and is equal to the magnitude of the particle's weight times its vertical displacement

Work and Energy

The Work of a Spring Force





If an elastic spring is elongated a distance ds,, then the work done by the force that acts on the attached particle is $dU = F_s ds = -ks\, ds$.

The work is negative since Fs acts in the opposite sense to ds

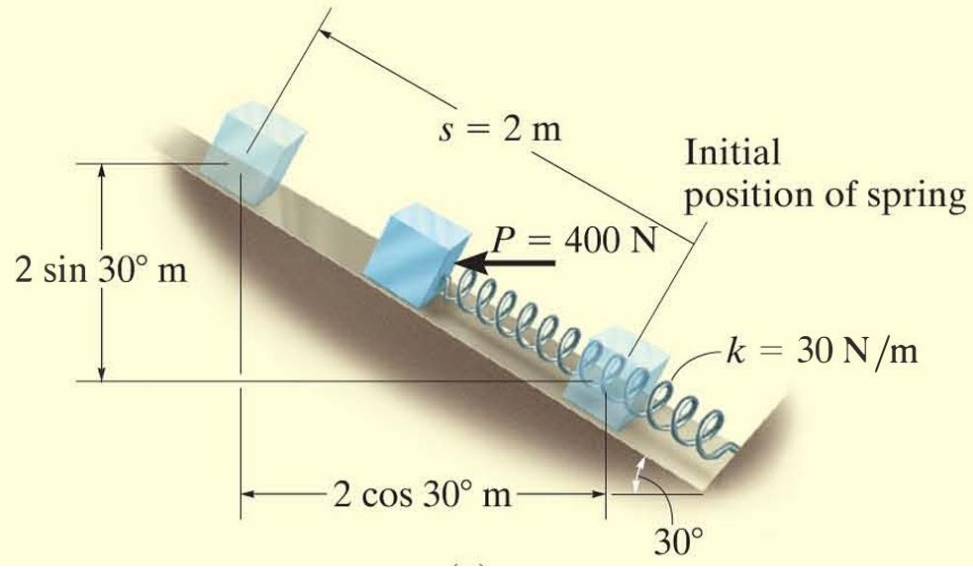
$$U_{1-2} = \int_{s_1}^{s_2} F_s ds = \int_{s_1}^{s_2} -ks\, ds$$

$$= -(\frac{1}{2} ks_2^2 - \frac{1}{2} ks_1^2)$$

This work represents the trapezoidal area under the line $F_s = ks$

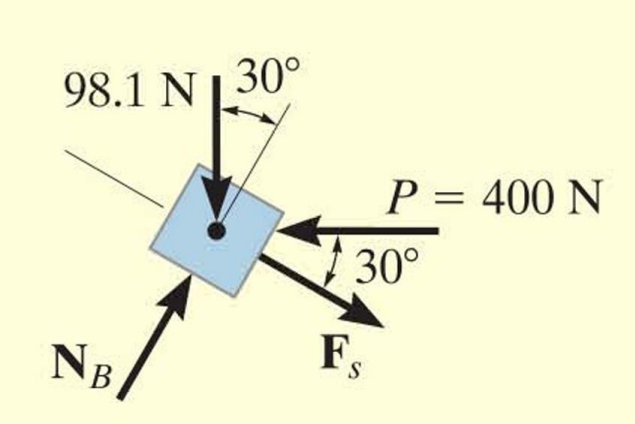
Example 1

The 10-kg block rests on the smooth incline. If the spring is originally stretched 0.5 m, determine the total work done by all the forces acting on the block when a horizontal force P = 400 N pushes the block up the plane s = 2 m.



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Work of P;

Since this force is constant, the work is determined using,

$$U_p = 400\,(2\cos 30) = 692.8\,J$$

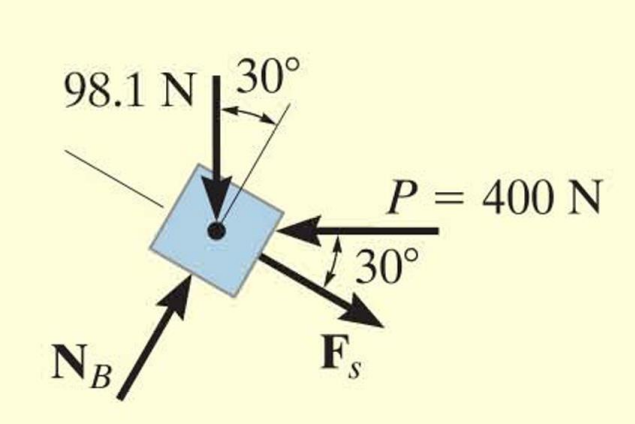
Work of Spring Force;

In the initial position the spring is stretched $s_1 = 0.5$ m and in the final position it is stretched $s_2 = 0.5\,m + 2\,m = 2.5\,m$. We require the work to be negative since the force and displacement are opposite to each other.

$$U_s = -\left[\frac{1}{2}(30)(2.5)^2 - \frac{1}{2}(30)(0.5)^2\right]J = -90\,J$$

Example 1

The 10-kg block rests on the smooth incline. If the spring is originally stretched 0.5 m, determine the total work done by all the forces acting on the block when a horizontal force P = 400 N pushes the block up the plane s = 2 m.



Work of Weight;

$$U_w = -(9.81)(2\sin 30) = -98.1\,J$$

Work of N_B ;

$$U_{N_B} = 0$$

Total Work ;

$$U_T = 692.8 - 90 - 98.1 = 505\,J$$

Principle of Work and Energy

For a system of forces $\boldsymbol{F_1}, \boldsymbol{F_2}, \boldsymbol{F_3}, \ldots, \boldsymbol{Fi}, \ldots, \boldsymbol{F_n}$ acting on a particle, the work done by the resultant $\boldsymbol{\Sigma F}$ of the system equals the sum of the scalar works done by each of its components.

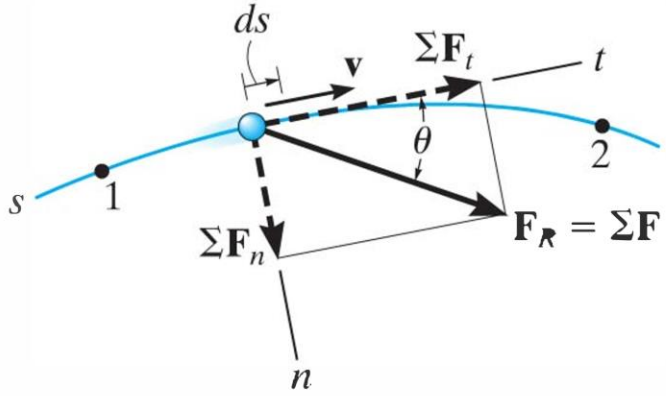
$$\boldsymbol{dU} = \boldsymbol{F_1}.d\boldsymbol{r} + \boldsymbol{F_2}.d\boldsymbol{r} + \boldsymbol{F_3}.d\boldsymbol{r} + \ldots + \boldsymbol{F_i}.d\boldsymbol{r} + \ldots . + \boldsymbol{F_n}.d\boldsymbol{r}$$
$$= \boldsymbol{\Sigma F}.d\boldsymbol{r}$$

$$U = \int \Sigma F.dr$$

Principle of Work and Energy

Substituting **Newton's second law of motion**

$$U = \int \Sigma F . dr = \int ma . dr = \int m\, a_t ds$$



From kinematics: $a = v\frac{dv}{ds}$

Therefore, $\sum U_{1-2} = \int_{v_1}^{v_2} mv\, dv = \frac{1}{2}m({v_2}^2 - {v_1}^2)$

This equation represents the principle of work and energy for the particle.
The term on the left is the sum of the work done by all the forces acting on the particle as the particle moves from point 1 to point 2.
The two terms on the right side, which are of the form $T = mv^2$, define the particle's final and initial kinetic energy, respectively.

Principle of Work and Energy

Kinetic Energy, T

The Kinetic Energy (T) of the particle is defined as

$$T = \frac{1}{2}mv^2$$

Like work, kinetic energy is a scalar

This is the total work which must be done on the particle to bring it from a state of rest to a velocity v

the work-energy equation for a particle is:

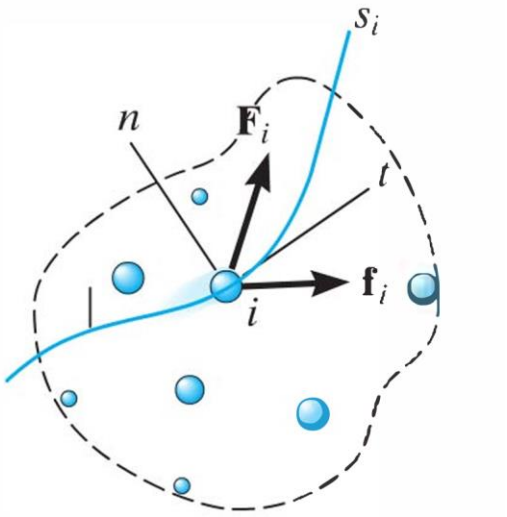
$$T_1 + \sum U_{1-2} = T_2$$

This equation involves only those forces which do work and give rise to changes in the magnitude of the velocities.

For a system of particles, the *work-energy equation* is applicable to the entire system.
U is the total or net work done on the system by forces external to it and ΔT is the change in the total kinetic energy of the system.

Principle of Work and Energy for a System of Particles

The principle of work and energy can be extended to include a system of particles isolated within an enclosed region of space



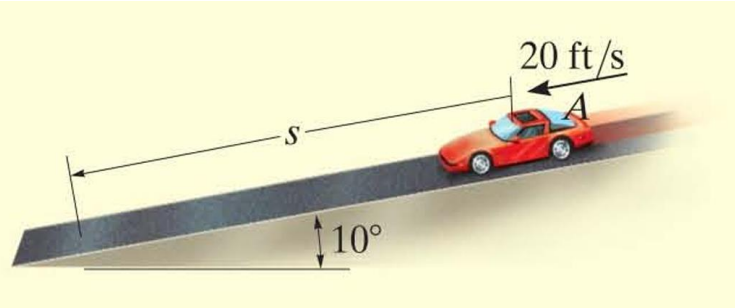
Inertial coordinate system

$$T_1 + \sum U_{1-2} = T_2$$

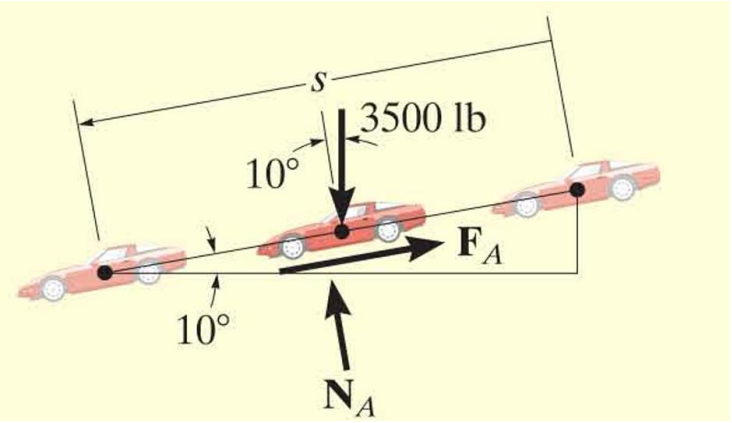
The initial kinetic energy of the system plus the work done by all the external acting on the system is equal to the final kinetic energy of the system.

Example 2

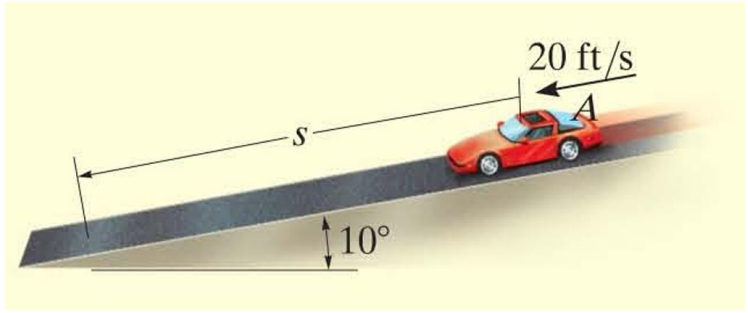
The 3500-lb automobile travels down the 10° inclined road at a speed of 20 ft/s. If the driver jams on the brakes, causing his wheels to lock, determine how far s the tires skid on the road. The coefficient of kinetic friction between the wheels and the road is $\mu_k = 0.5$.

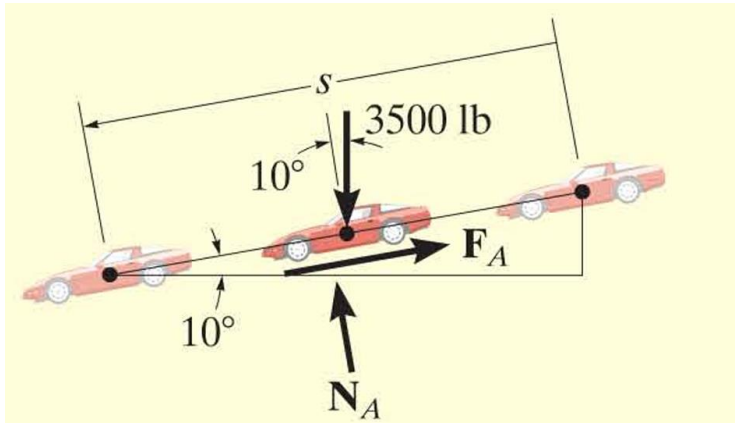


Example 2



Example 2





Work

The normal force NA does no work since it never undergoes displacement along its line of action.

The weight, 3500 lb, is displaced s sin 10° and does positive work.

The frictional force FA does work when it undergoes a displacement s. This work is negative since it is in the opposite sense of direction to the displacement.

$$\sum F_n = 0$$

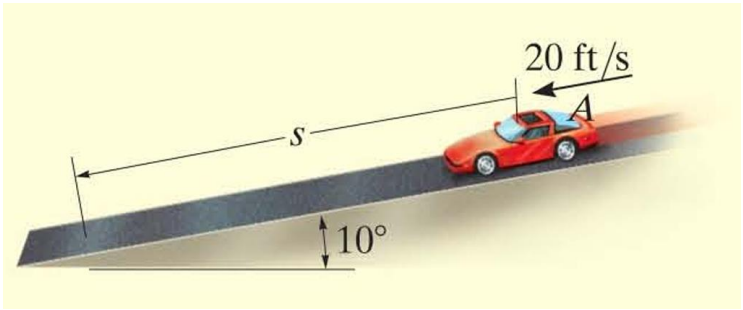
$$N_A - 3500 \cos 10 = 0$$

$$N_A = 3446.8\ lb$$

Therefore,

$$F_A = \mu_k N_A = 0.5\ (3446.8) = 1723.4\ lb$$

Example 2



Principle of Work and Energy

$$T_1 + \sum U_{1-2} = T_2$$

$$\left(\frac{1}{2}\right)\left(\frac{3500\ lb}{32.2\ fts^{-2}}\right)(20\ fts^{-2}) + 3500\ (s \sin 10)\ - (1723.4)s = 0$$

$$s = \ 19.5\ ft$$

Example 3

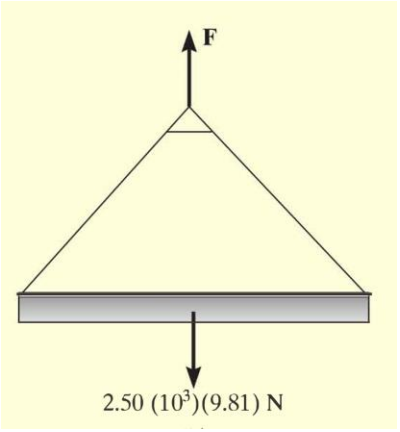
For a short time the crane lifts the 2.50-Mg beam with a force of $F = (28 + 3s^2)\ kN$. Determine the speed of the beam when it has risen s = 3 m. Also, how much time does it take to attain this height starting from rest?



Example 3

For a short time the crane lifts the 2.50-Mg beam with a force of $F = (28 + 3s^2)\ kN$. Determine the speed of the beam when it has risen s = 3 m. Also, how much time does it take to attain this height starting from rest?

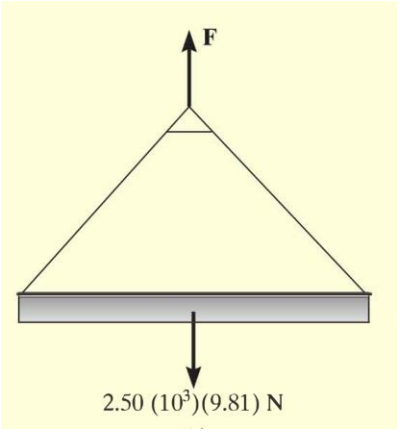




Example 3

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Principle of Work and Energy

$$T_1 + \sum U_{1-2} = T_2$$

$$0 + \int_0^s (28 + 3s^2)10^3\ ds\ - (2.50)(10^3)(9.81)s = \frac{1}{2}(2.5)(10^3)v^2$$

$$28(10^3)s + 10^3 s^3\ - 24.525(10^3)s = 1.25\ (10^3)v^2$$

$$v = (2.78s + 0.8\ s^3)^{1/2}$$

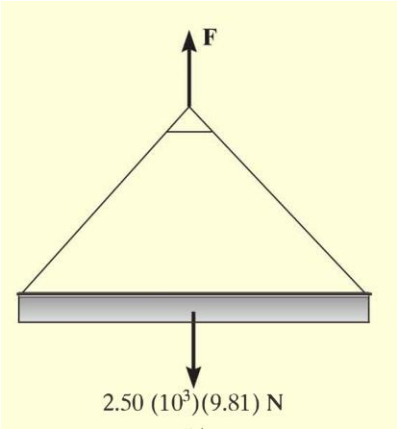
When s =3;

$$v = 5.47\ ms^{-1}$$

Example 3

For a short time the crane lifts the 2.50-Mg beam with a force of $F = (28 + 3s^2)\ kN$. Determine the speed of the beam when it has risen s = 3 m. Also, how much time does it take to attain this height starting from rest?





Kinematics

$$v = \frac{ds}{dt}$$

$$(2.78s + 0.8\,s^3)^{1/2} = \frac{ds}{dt}$$

$$dt = \frac{ds}{(2.78s + 0.8\,s^3)^{1/2}}$$

$$t = \int_0^3 \frac{ds}{(2.78s + 0.8\,s^3)^{1/2}}$$

$$t = 1.79\ s$$

Power and Efficiency

Power

The term "power" provides a useful basis for choosing the type of motor or machine which is required to do a certain amount of work in a given time.

The power generated by a machine or engine that performs an amount of work dU within the time interval dt is therefore

$$P = \frac{dU}{dt}$$

$$P = \frac{dU}{dt} = F\frac{dr}{dt} = F.v$$

$$P = F.v$$

Hence, power is a scalar, where in this formulation v represents the velocity of the particle which is acted upon by the force F.

Power and Efficiency

Units of Power

The basic units of power used in the SI and FPS systems are the watt (W) and horsepower (hp) respectively.

These units are defined as,

$$1W = 1\,J/s = 1\,N.m/s$$

$$1\,hp = 550\,ft.lb/s$$

$$1\,hp = 746\,W$$

Power and Efficiency

Efficiency

The mechanical efficiency of a machine is defined as the ratio of the output of useful power produced by the machine to the input of power supplied to the machine.

$$\eta = \frac{Power\ output}{Power\ \ input}$$

If energy supplied to the machine occurs during the same time interval at which it is drawn, then,

$$\eta = \frac{energy\ output}{energy\ \ input}$$

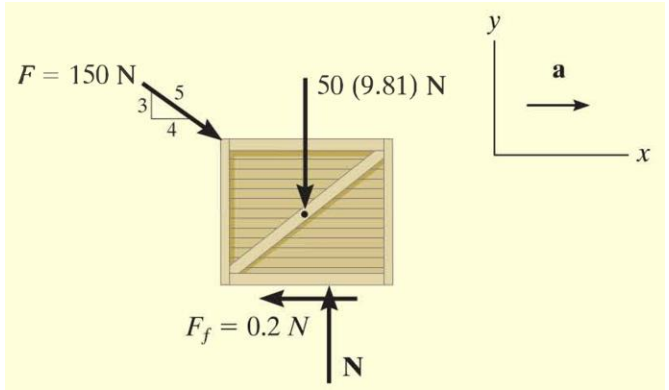
Example 4

The man in Figure pushes on the 50-kg crate with a force of F = 150 N. Determine the power supplied by the man when t = 4 s. The coefficient of kinetic friction between the floor and the crate is $\mu_k = 0.2$. Initially the crate is at rest.



Example 4

The man in Figure pushes on the 50-kg crate with a force of F = 150 N. Determine the power supplied by the man when t = 4 s. The coefficient of kinetic friction between the floor and the crate is $\mu_k = 0.2$. Initially the crate is at rest.



To determine the power developed by the man, the velocity of the 150-N force must be obtained first. Applying the equation of motion,

Equation of Motion:

$$\uparrow \sum F_y = ma_y$$

$$N_c - \left(\frac{3}{5}\right)150 - 50(9.81) = 0$$

$$N = 580.5\,N$$

$$\rightarrow \sum F_x = ma_x$$

$$\left(\frac{4}{5}\right)150 - 0.2(580.5) = 50\,a$$

$$a = 0.078\,ms^{-2}$$

Example 4

The man in Figure pushes on the 50-kg crate with a force of F = 150 N. Determine the power supplied by the man when t = 4 s. The coefficient of kinetic friction between the floor and the crate is $\mu_k = 0.2$. Initially the crate is at rest.

The velocity of the crate when $t = 4s$ is therefore,

$$\rightarrow v = v_0 + a_c t$$

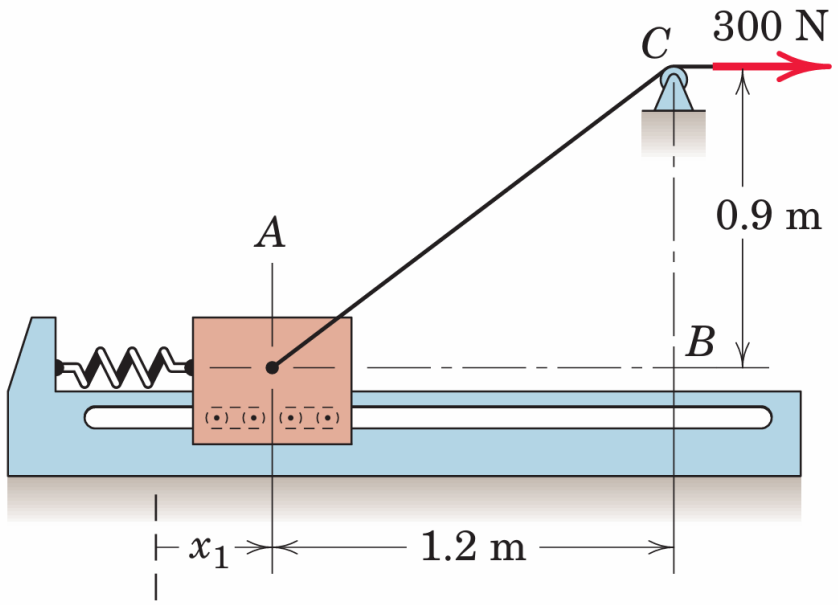
$$v = 0 + 0.078(4) = 0.312\, ms^{-1}$$

The power supplied to the crate by the man when t = 4 s is therefore,

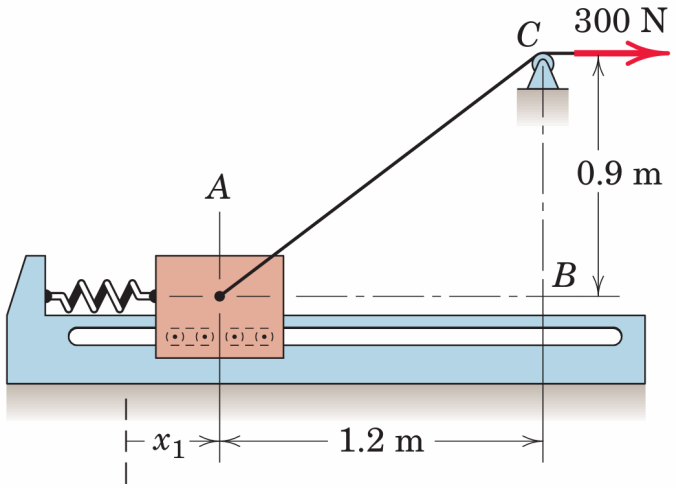
$$p = F.v = F_x v = \left(\frac{4}{5}\right)(150)(0.312) = 37.4\, W$$

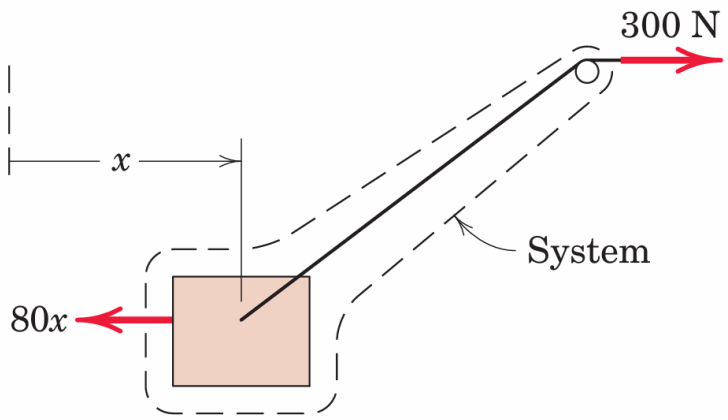
Question 1

The 50-kg block at A is mounted on rollers so that it moves along the fixed horizontal rail with negligible friction under the action of the constant 300-N force in the cable. The block is released from rest at A, with the spring to which it is attached extended an initial amount 0.233 m. The spring has a stiffness k 80 N/m. Calculate the velocity v of the block as it reaches position B.



Question 1





Solution. It will be assumed initially that the stiffness of the spring is small enough to allow the block to reach position B. The active-force diagram for the system composed of both block and cable is shown for a general position. The spring force $80x$ and the 300-N tension are the only forces external to this system which do work on the system. The force exerted on the block by the rail, the weight of the block, and the reaction of the small pulley on the cable do no work on the system and are not included on the active-force diagram.

As the block moves from $x_1 = 0.233$ m to $x_2 = 0.233 + 1.2 = 1.433$ m, the work done by the spring force acting on the block is

$$[U_{1\text{-}2} = \tfrac{1}{2}k(x_1^2 - x_2^2)] \quad U_{1\text{-}2} = \tfrac{1}{2}80[0.233^2 - (0.233 + 1.2)^2]$$

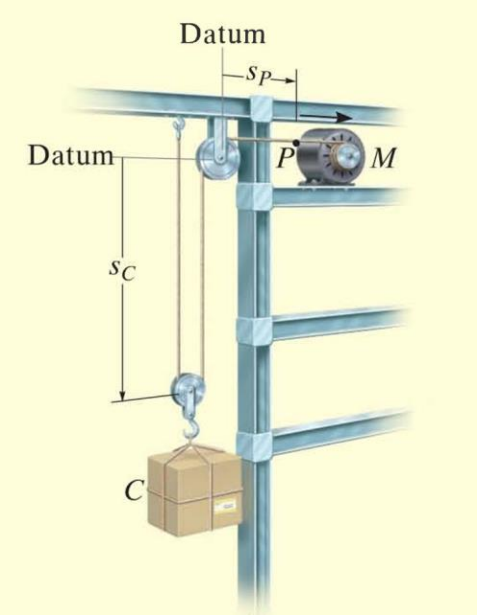
$$= -80.0 \text{ J}$$

The work done on the system by the constant 300-N force in the cable is the force times the net horizontal movement of the cable over pulley C, which is $\sqrt{(1.2)^2 + (0.9)^2} - 0.9 = 0.6$ m. Thus, the work done is $300(0.6) = 180$ J. We now apply the work-energy equation to the system and get

$$[T_1 + U_{1\text{-}2} = T_2] \quad 0 - 80.0 + 180 = \tfrac{1}{2}(50)v^2 \qquad v = 2.00 \text{ m/s} \qquad \textit{Ans.}$$

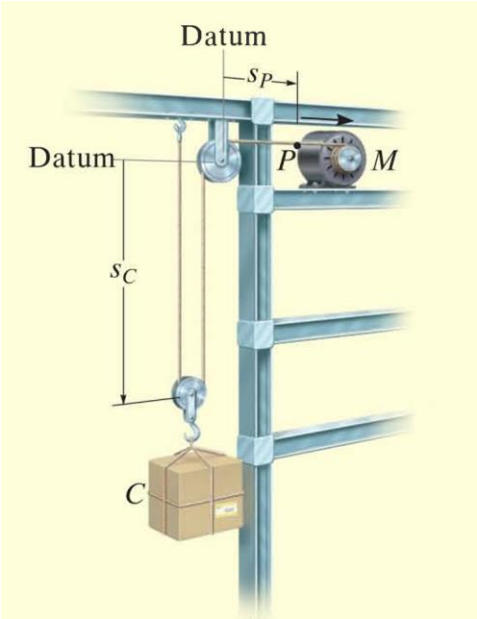
Question 2

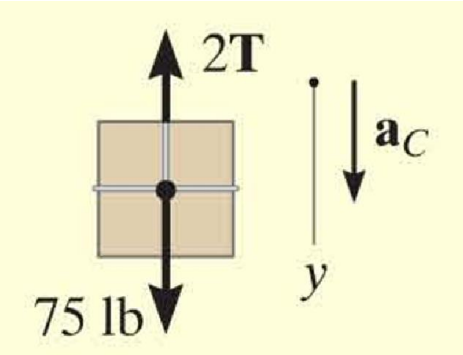
The motor M of the hoist shown in figure lifts the 75-lb crate C so that the acceleration of point P is $4\ ft/S^2$. Determine the power that must be supplied to the motor at the instant P has a velocity of 2 ft/s. Neglect the mass of the pulley and cable and take $\eta = 0.85$.



Question 2

The motor M of the hoist shown in figure lifts the 75-lb crate C so that the acceleration of point P is $4\ ft/S^2$. Determine the power that must be supplied to the motor at the instant P has a velocity of 2 ft/s. Neglect the mass of the pulley and cable and take $\eta = 0.85$.





SOLUTION

In order to find the power output of the motor, it is first necessary to determine the tension in the cable since this force is developed by the motor.

From the free-body diagram, Fig. 14–16*b*, we have

$$+\downarrow \quad \Sigma F_y = ma_y; \qquad -2T + 75\ \text{lb} = \frac{75\ \text{lb}}{32.2\ \text{ft/s}^2} a_c \qquad (1)$$

The acceleration of the crate can be obtained by using kinematics to relate it to the known acceleration of point P, Fig. 14–16*a*. Using the methods of Sec. 12.9, the coordinates s_C and s_P can be related to a constant portion of cable length l which is changing in the vertical and horizontal directions. We have $2s_C + s_P = l$. Taking the second time derivative of this equation yields

$$2a_C = -a_P \qquad (2)$$

Since $a_P = +4\ \text{ft/s}^2$, then $a_C = -(4\ \text{ft/s}^2)/2 = -2\ \text{ft/s}^2$. What does the negative sign indicate? Substituting this result into Eq. 1 and *retaining* the negative sign since the acceleration in *both* Eq. 1 and Eq. 2 was considered positive downward, we have

$$-2T + 75\ \text{lb} = \left(\frac{75\ \text{lb}}{32.2\ \text{ft/s}^2}\right)(-2\ \text{ft/s}^2)$$

$$T = 39.83\ \text{lb}$$

The power output, measured in units of horsepower, required to draw the cable in at a rate of 2 ft/s is therefore

$$P = \mathbf{T} \cdot \mathbf{v} = (39.83\ \text{lb})(2\ \text{ft/s})[1\ \text{hp}/(550\ \text{ft} \cdot \text{lb/s})]$$

$$= 0.1448\ \text{hp}$$

This *power output* requires that the motor provide a *power input* of

$$\text{power input} = \frac{1}{\epsilon}(\text{power output})$$

$$= \frac{1}{0.85}(0.1448\ \text{hp}) = 0.170\ \text{hp} \qquad \textit{Ans.}$$